

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

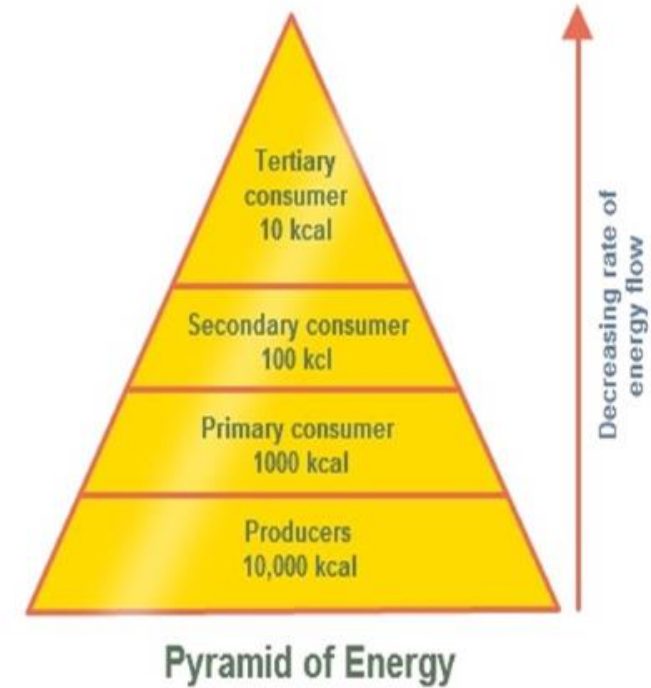
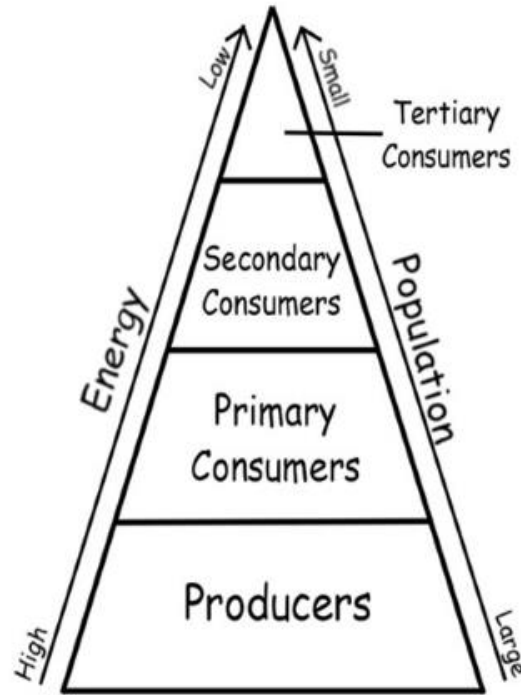
A COMPREHENSIVE COVERAGE

MODULE - 28

WATER POLLUTION - 5

BIO-MAGNIFICATION AND BIO-ACCUMULATION

- **Definition** - Trophic Levels – Trophic levels are the representation of various energy levels in an ecosystem.
- The trophic level of an organism is the position it occupies in a food chain.

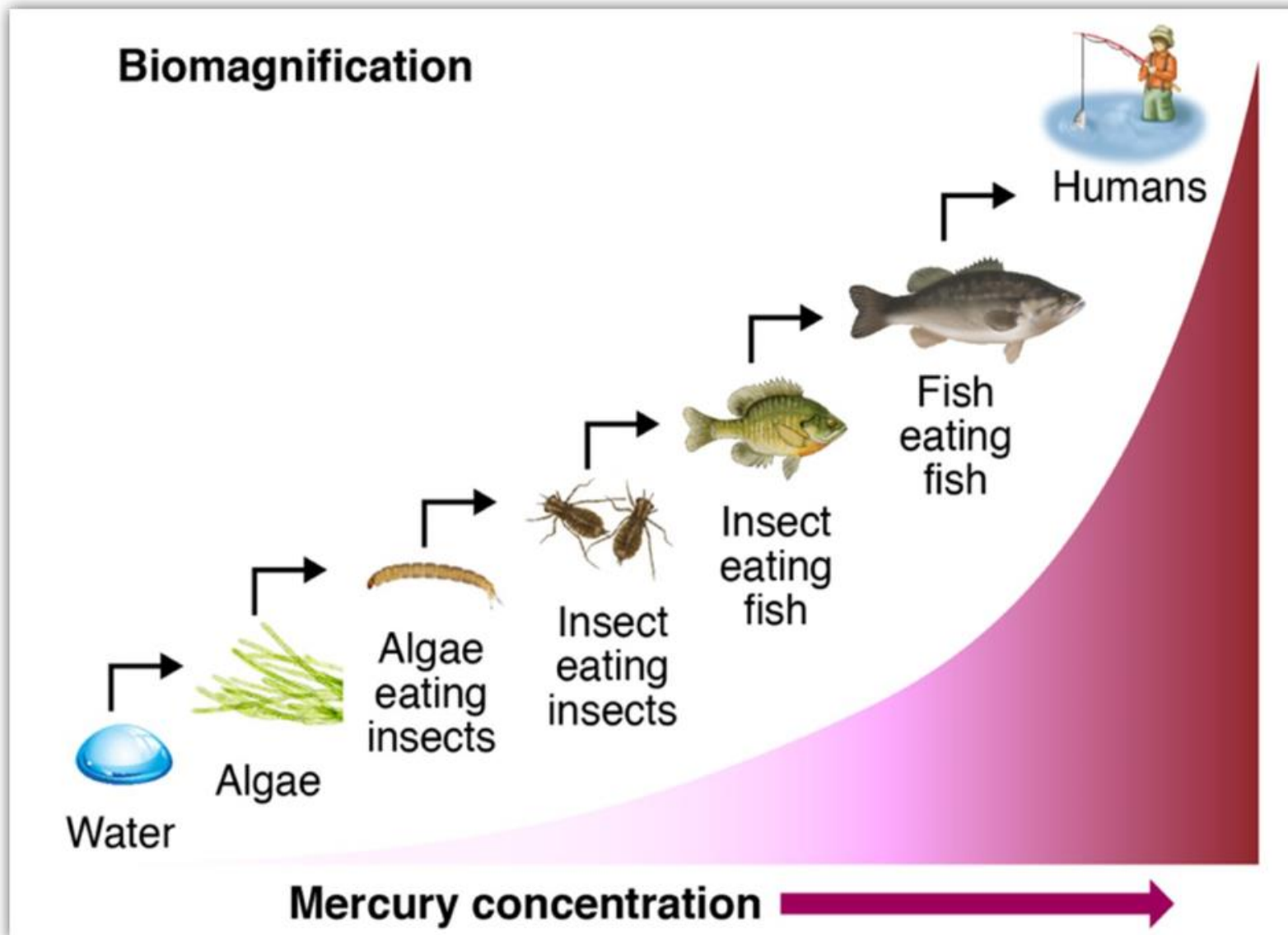


BIO-MAGNIFICATION

- It refers to the tendency of pollutants to concentrate as they move from one trophic level to the higher one.
- There is an increase in concentration of pollutants as they move from one link in a food chain to another.
- In order for bio-magnification to occur, the pollutant must be long lived, mobile and should be soluble in fat.



BIO-MAGNIFICATION

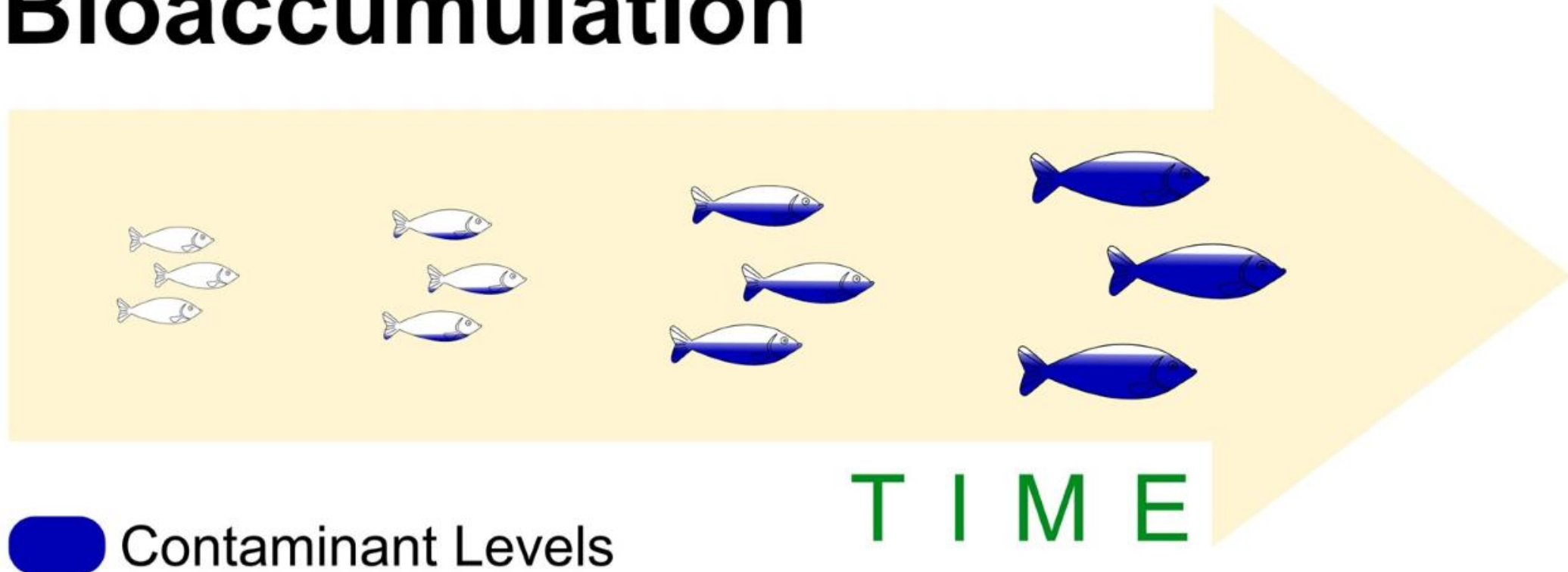


*Bio-magnification
is nature's way of
dealing karma!*

BIO- ACCUMULATION

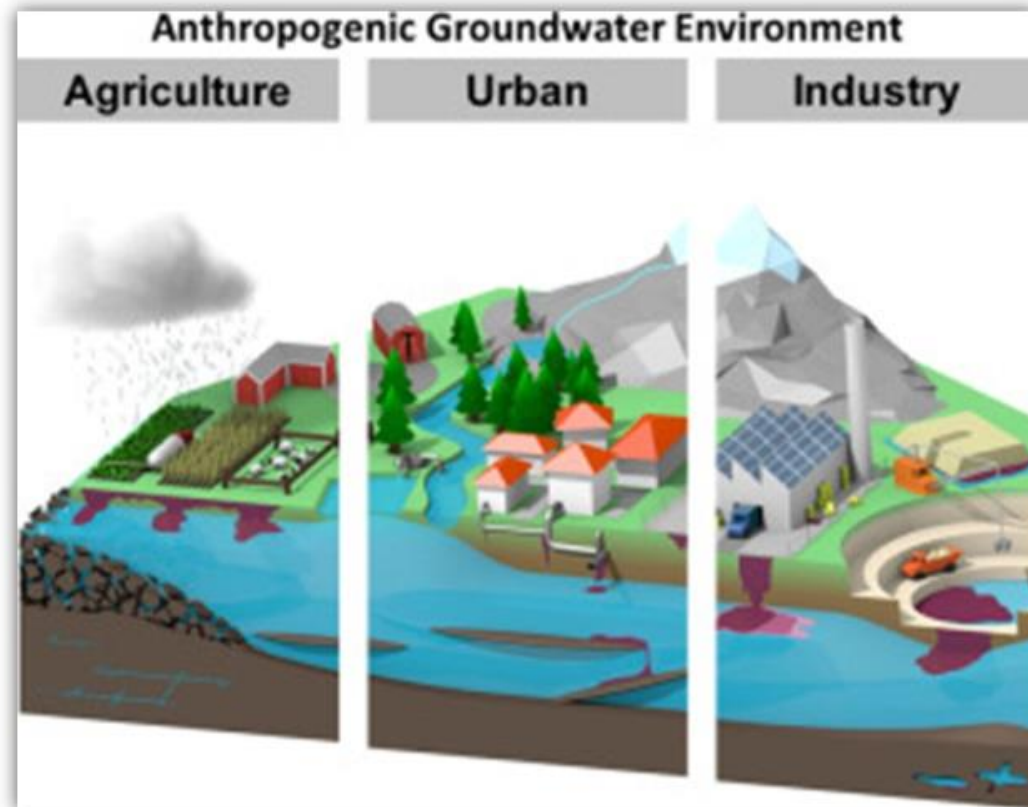
- It refers to the process of pollutants entering a food chain from the environment.
- Here, there is an increase in concentration of pollutants as they move from the environment to the organism in a food chain.

Bioaccumulation



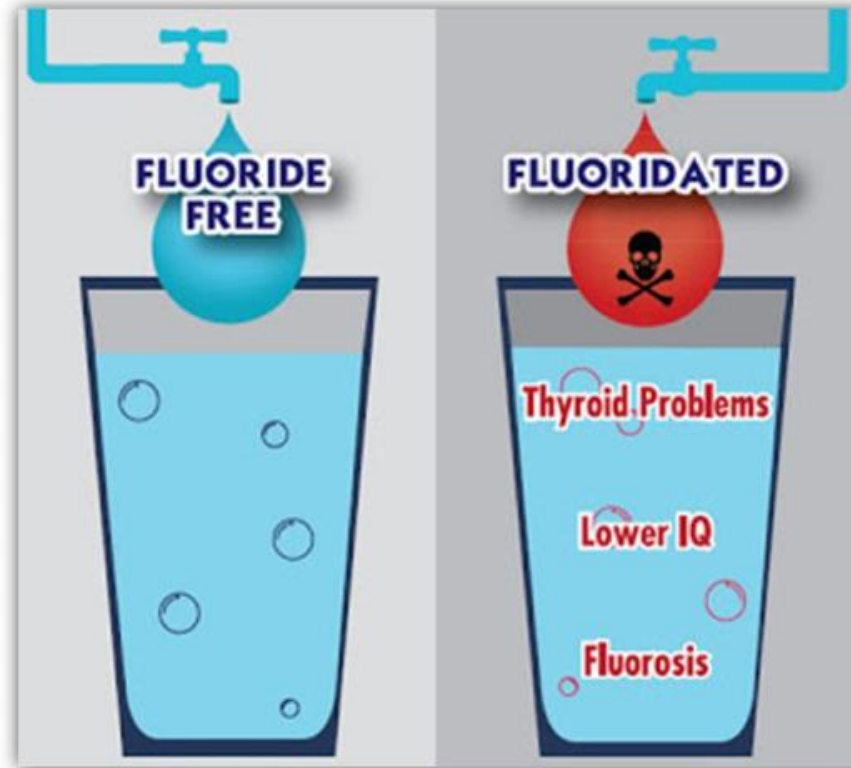
GROUNDWATER

- Groundwater accounts for nearly 80% of the rural domestic water need and 50% of the urban water need in India. The percentage of dependence is increasing every day.
- Unlike surface water which receives a lot of attention because of visible pollution, polluted groundwater keeps affecting public health slowly and silently.
- Deterioration of groundwater is caused by both Geogenic and Anthropogenic causes.
- Geogenic pollution of groundwater refers to naturally occurring elevated concentrations of certain elements in the groundwater.
- Geogenic contaminants originate from the rock-water interaction.
- The major Geogenic contaminants in India are: fluoride, arsenic, salinity and iron.



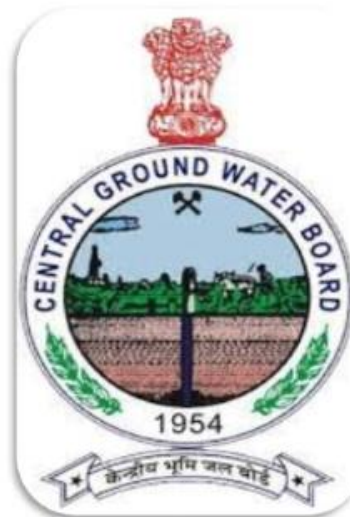
GROUNDWATER - FLUORIDE

- Fluoride concentration exceeding 1.5ppm for drinking water is widely reported across the length and breadth of the country.
- It is most prevalent in Telangana, AP, TN, UP, Gujarat and Rajasthan where many districts have reported excess levels of fluoride in groundwater.



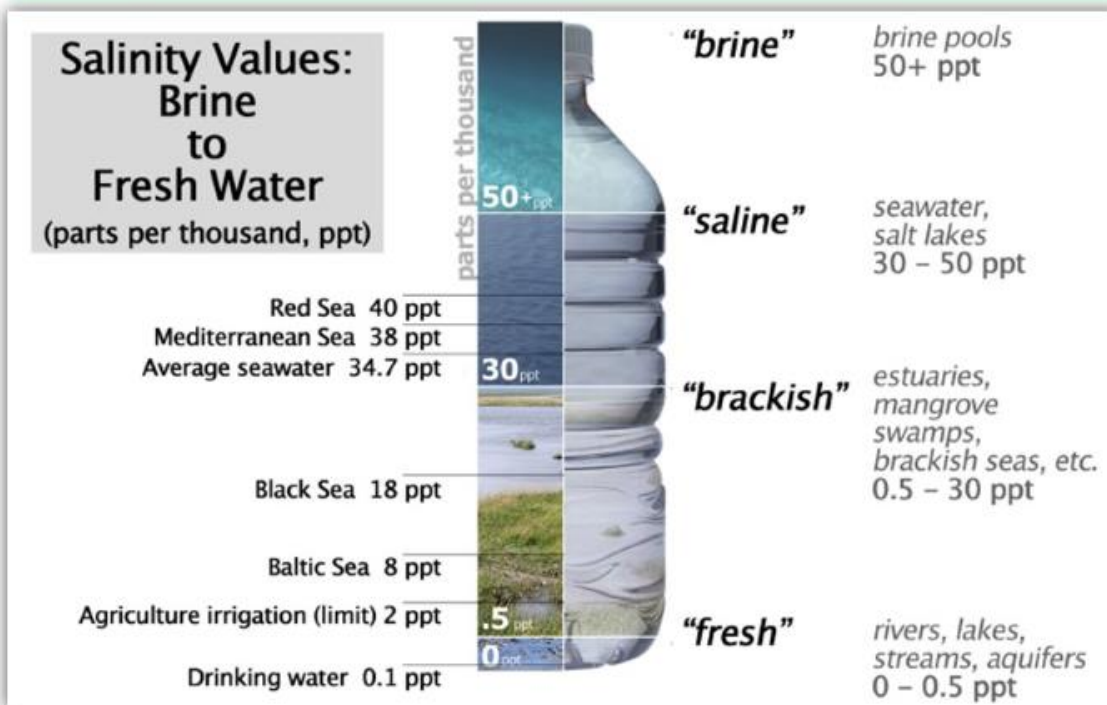
GROUNDWATER - ARSENIC

- As per BIS standards, the maximum permissible limit of arsenic in drinking water was 50 PPB, which was revised to 10 ppb in 2015.
- According to studies conducted by CGWB, arsenic concentration in excess of 10 PPB has been reported from parts of 21 states.
- However, most of the contamination is mostly confined to the Indo-Ganga-Brahmaputra alluvial terrain, covering parts of WB, UP, Bihar, Punjab, Haryana, Assam and also Manipur.



GROUNDWATER - SALINITY

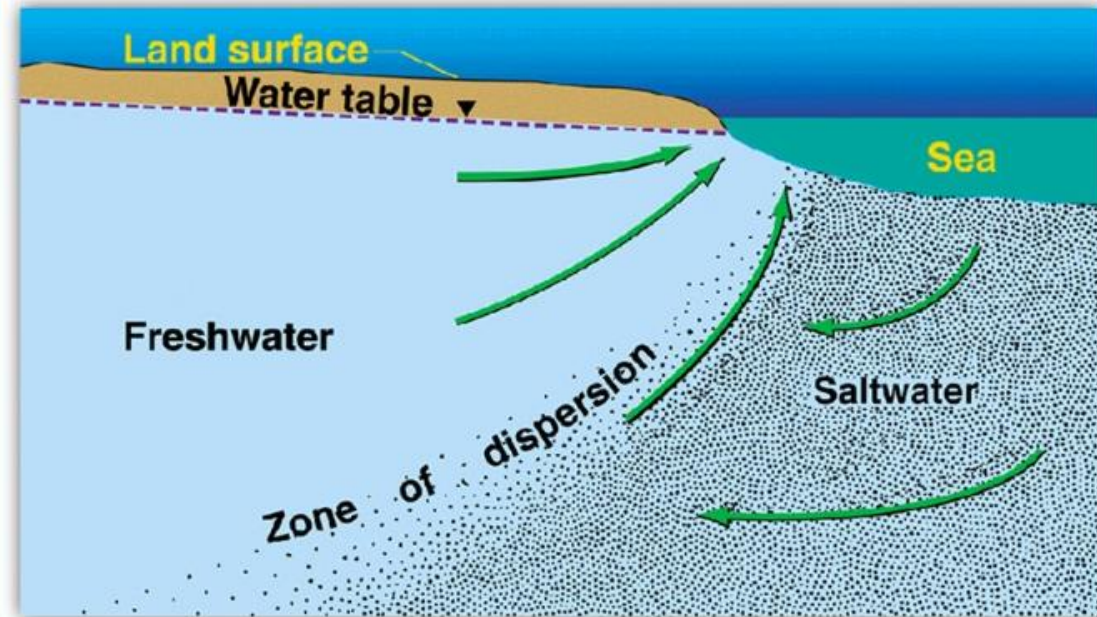
- Salinity does not cause serious health effects as compared to arsenic and fluoride.
- Salinity could be inland, caused by rock-water interaction or coastal salinity, caused by seawater intrusion.
- Inland salinity has been reported mostly from Rajasthan, Gujarat and Haryana.
- Seawater intrusion has been reported from the coastal areas of TN, WB, Odisha, Gujarat and AP.



GROUNDWATER

SALINITY

- **Seawater Intrusion:** Is the movement of seawater into freshwater aquifers. It is caused by decrease in groundwater levels and rise in sea levels.
- Because saline water has a higher mineral content than freshwater, it is denser and has higher pressure, so salt water can push inland beneath the freshwater.
- Groundwater pumping from coastal areas has increased salt water intrusion in many coastal districts.
- Water extraction drops the level of fresh groundwater, reducing its pressure and allowing salt water to flow further inland. So, groundwater extraction is the primary cause of seawater intrusion.



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